MIET2572 – Wearable Tech Devices Workshop #4

Week 9: From Signal to Metrics

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Objective

In this workshop you will move from event detection to movement metrics.

Key idea: Peaks tell you when something happened. Metrics help define what that movement means.

Week 3: detect events → Week 9: define movement quality

Expectation
This is an exploratory session.
Your goal is not to build a perfect model.
Your goal is to understand what your signal represents.

A smartphone does not measure fatigue, smoothness, or stability.
It measures signals.
Engineering begins when we decide what those signals mean.

Review – What We Have Seen So Far

Body → Wearable → Signal → Model → Metric

- Peak detection
- Thresholding
- Smoothing
- Temporal constraints
- Model assumptions
- Failure modes

Context Matters

Metrics do not exist in isolation.

The same metric can change depending on the body, task, sensor placement, environment, and experimental condition.

Examples

- sensor placement
- task constraints
- fatigue or recent physical activity
- injury, fasting vs fed state, pain or discomfort
- surface conditions and inclination
- object weight
- dominant vs non-dominant hand
- eyes open vs eyes closed
- cognitive distraction or time pressure
- external rhythm/music
- environmental conditions

A metric is only meaningful when the context is defined.

Task 1 – From Events to Metrics

Action
- Identify one quantity in your project that describes movement quality

Examples

- Cadence
- Stability
- Smoothness
- Variability
- Fatigue
- Symmetry

Question

- What aspect of movement are you trying to quantify?

Observation

Real wearable recordings may contain irrelevant regions before or after the movement.

Selecting the analysis window is part of the preprocessing stage. This is known as **ROI (Region of Interest).**

Task 2 – RMS and Signal Intensity

RMS measures signal intensity, not events.

Action
- Compute RMS for one movement trial
- Compare RMS across two different movement conditions

Project Relevance

- Tremor
- Impact
- Instability
- Fatigue

Analyse

- Does RMS increase with movement intensity?

Task 3 – Standard Deviation and Variability

STD measures variability and consistency.

Action
- Compute the standard deviation of your signal
- Compare two movement conditions

Project Relevance

- Variability
- Stability
- Rhythm consistency
- Movement repeatability

Question

- What type of movement produces larger variability?
- Is the STD caused by instability, inconsistency, intentional movement changes, or the context of the task?
- Does STD reflect instability, inconsistency, or intentional movement changes?

Task 4 – Jerk and Smoothness

Jerk measures abruptness of motion.

Signal	Meaning
Position	where
Velocity	how fast
Acceleration	how strongly speed changes
Jerk	how abruptly acceleration changes

Acceleration tells you intensity → Jerk tells you smoothness.

Action
- Compute jerk from acceleration data
- Compare controlled vs unstable movement

Observation: Jerk estimation is highly sensitive to preprocessing choices.

raw acceleration
↓
smoothing
(parameter selection matters)
↓
derivative (Jerk)

Project Relevance

- Smoothness
- Coordination
- Throwing
- Movement quality

Question

- Which condition produces larger jerk?
- Does larger jerk reflect reduced smoothness, a change in task context, or noise amplification?

Task 5 – Peak Interval and Cadence

Sometimes timing matters more than amplitude.

Action
- Detect peaks
- Compute time intervals between peaks

Project Relevance

- Cycling
- Tapping
- Rhythm
- Gait

Analyse

- Is cadence stable across trials?

Task 6 – Apply Metrics to Your Project. (To be also continued on Week 11)

Action
- Select one metric relevant to your project
- Apply it to your pilot data

Analyse

- Does the metric capture meaningful behaviour?
- Does the metric respond to changes in movement?

Metrics are models.

They only work under assumptions.

Results Log

Metric	Observation	Interpretation
RMS		
STD		
Jerk		
Cadence		

Final Reflection

- Which metric was most useful for your project?
- Which metric was most sensitive to noise?
- What assumptions define your metric?

Key Takeaway

Peaks tell you when something happened.

Metrics transform signals into interpretation.

A signal only becomes useful when we define what it means.

Looking Ahead – From Metrics to Indices

An index measures an idea.

Examples

- BMI
- HRV score
- Fatigue index

Metrics measure quantities.

Indices define meaning.

An index combines multiple assumptions into a single interpretation.

Possible Examples

Stability Index:

$$SI = \frac{1}{STD}$$

Smoothness Index:

$$SM = \frac{1}{RMS(Jerk)}$$

Fatigue Index:

$$FI = \frac{Peak_{initial} - Peak_{final}}{Peak_{initial}}$$

A wearable does not only measure signals.

It makes claims about the body.

A claim depends on how signals are interpreted.

Different models can produce different claims from the same signal.

Question

- What assumptions define your index?
- Could a different *wearable designer* define the same claim differently?
- If your index does not exist yet, can you invent one?

This will be explored in a future workshop.